# HVAC Power Forecasting Service

This document outlines the architecture, methodology, and mathematical models behind our Day-Ahead HVAC Forecasting Service. It is designed to predict the power consumption of three specific HVAC units at 10-minute intervals, ensuring high reliability through a cascading fallback strategy.

## 1. System Overview

The forecasting service is a robust, automated pipeline designed to run in a serverless environment (like AWS Lambda). Its primary job is to look at tomorrow's weather forecast, combine it with historical energy usage, and predict exactly how much power the HVAC systems will draw every 10 minutes throughout the day.

**Key Objectives:**

- **Granular Tracking:** Predicts power usage in 10-minute blocks (144 predictions per day, per meter).
- **Fault Tolerance:** Uses a 3-tier "cascade" system to ensure a forecast is always generated, even if machine learning models fail or data is missing.
- **Building Physics:** Incorporates thermal inertia (how the building retains heat) rather than just looking at the current outside temperature.

## 2. Feature Engineering

Before any forecasting happens, the raw data is transformed to help the models understand time, physics, and human behavior.

### A. Diurnal Weather Generation

Standard weather forecasts usually only provide daily minimum and maximum temperatures. Since we need 10-minute granularity, the system mathematically stretches these two data points into a continuous, realistic daily temperature curve (a sine wave) that peaks in the afternoon and drops at night.

### B. Building Thermal Inertia

A building doesn't instantly heat up or cool down when the outside temperature changes; it absorbs and releases heat slowly. The code calculates **1-hour and 3-hour rolling averages** of the outside temperature (`temp_roll_1h`, `temp_roll_3h`). This teaches the model about the

building's thermal mass—knowing that a 30°C afternoon following a cool morning requires different cooling power than a 30°C afternoon during a 3-day heatwave.

### C. Human & Operational Context

- **Cyclic Time:** Time of day is converted into a circle using sine and cosine waves. This tells the model that 23:50 and 00:00 are only 10 minutes apart, rather than at opposite ends of a number line.
- **Mall Schedule:** A binary flag (`is_mall_active`) indicates when the Building Management System (BMS) turns the system on (07:30 to 23:00).
- **Calendar Effects:** The system flags weekends and queries a live registry of Portuguese public holidays, as occupancy (and therefore cooling load) drops significantly on these days.
- **Timezone Precision:** Weather curves and human schedules are calculated in local time (`Europe/Lisbon`), while database storage remains strictly in `UTC` to prevent daylight saving time data collisions.

## 3. The 3-Tier Forecasting Cascade

To guarantee that downstream systems always receive a forecast, the service uses a highly resilient 3-Tier strategy. It evaluates the most advanced method first and gracefully downgrades if safety checks fail.

### Tier 1: LightGBM Machine Learning (The Primary Engine)

- **How it works:** Uses `LightGBM`, an advanced, high-performance machine learning model. It looks at all engineered features (weather, time, inertia, schedules) to find complex, non-linear patterns.
- **Safety Check:** The model splits the data, training on older data and testing itself on the most recent 7 days. It calculates an accuracy score ($R^2$).
- **Validation:** If the model achieves an $R^2 \geq 0.65$ (meaning it successfully explains 65% or more of the variance in recent data) and has at least 1,000 historical data points, it is approved.
- **Safety Cap:** Predictions are strictly capped at 115% of the highest historically observed power draw to prevent runaway ML predictions.

### Tier 2: Same-Weekday Historical Median (The Safe Fallback)

- **How it works:** If the ML model is confused by anomalies (e.g., erratic recent data dropping the $R^2$ below 0.65), the system falls back to historical averages.
- **Method:** It looks at the target day (e.g., a Tuesday) and calculates the median power consumption for every specific 10-minute slot across all past Tuesdays.
- **Why Median?** The median ignores extreme outliers (like a Tuesday where the sensor broke or a rare heatwave occurred), providing a highly stable, "typical" profile.

### Tier 3: Naive Persistence (The Emergency Parachute)

- **How it works:** If Tier 2 fails (e.g., a brand new meter with no historical "Tuesdays" on record), the system looks at the exact power curve from yesterday.
- **Method:** "Whatever happened yesterday at 14:00 will happen today at 14:00." It provides a rudimentary baseline rather than crashing the system.
- **Failsafe:** If there isn't even data for yesterday, it outputs flat zeros.

# 4. Key Mathematical Formulas

For technical implementations, here are the core formulas driving the feature engineering:

**Diurnal Temperature Curve:** To estimate the temperature at any specific minute of the day based only on the daily Minimum ($T_{min}$) and Maximum ($T_{max}$):

$$T_{mean} = \frac{T_{max} + T_{min}}{2}$$

$$T_{amp} = \frac{T_{max} - T_{min}}{2}$$

$$T(h) = T_{mean} + T_{amp} \cdot \sin\left(2\pi \frac{h-9}{24}\right)$$

*(Where $h$ is the local hour in decimal format, e.g., 14:30 = 14.5. The - 9 shifts the peak temperature to mid-afternoon, around 15:00).*

**Cyclic Time Encoding:** To represent 144 ten-minute slots ($S$) continuously:

$$Time_{sin} = \sin\left(2\pi\frac{S}{144}\right)$$

$$Time_{cos} = \cos\left(2\pi\frac{S}{144}\right)$$

# 5. Database & Cloud Architecture

- **Stateless Execution:** Designed as a Lambda function (serverless service), it spins up, does the heavy lifting in memory using Pandas/Numpy, writes to the database, and shuts down.

- **Bulk Database Upserts:** Uses `psycopg2.extras.execute_values` to package all 432 data points (144 slots × 3 meters) into a single, highly efficient PostgreSQL transaction.

- **Conflict Resolution:** Utilizes `ON CONFLICT DO UPDATE`. If a forecast was already generated for tomorrow (e.g., manually triggered early), running it again safely overwrites the old predictions with the latest weather insights.